Haider Ali

hali.msai22seecs@seecs.edu.pk | +92 344 8388121 | LinkedIn | GitHub | Portfolio | Google Scholar

Research Interests

Computer Vision, Image Processing, Generative Diffusion Models, Uncertainty-Aware Vision, and Real-Time Recognition.

Education

National University of Sciences and Technology (NUST), Pakistan, MS in Artificial Intelligence (CGPA: 3.55/4.0) Aug 2022 – Apr 2025

Relevant Courses: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Reinforcement Learning, Artificial Intelligence, Mathematical Methods for Artificial Intelligence
Thesis: Offline Handwritten Signature Generation Using Generative Models (GPA: 4.0/4.0)
Supervisor: Prof. Dr. Faisal Shafait

Balochistan University of IT, Engineering and Management Sciences (BUITEMS), Pakistan, BS in Software Engineering (CGPA: 3.92/4.0) Aug 2017 – Dec 2021

Dean's Honor List

English Proficiency: IELTS Academic, overall band 7.0

Publications

Haider Ali, Maxim Dolgich, Ulrich Schwanecke, Adrian Ulges, Faisal Shafait. **Non-local sparse attention based multi-image super resolution**. [Accepted IGARSS 2026 (Poster), Washington, D.C., USA] [Paper][Code]

Haider Ali, Momina Moetesum, Adnan Ul-Hasan, Faisal Shafait, M. Imran Malik. **GeoCT-SigNet: A Geometry-Aware Hybrid CNN–Transformer Network for Offline Signature Verification**. [Under review in IEEE Access 2026] [Preprint][Code]

Haider Ali, Fabian Stahl, Maxim Dolgich, Ulrich Schwanecke, Adrian Ulges, Faisal Shafait. **RGB Super-Resolution for Multispectral Remote Sensing Images**. [Accepted IGARSS 2025 (Poster), Brisbane, Australia][Paper][Code]

Industry Experience

Machine Learning Engineer, Dcube (US-based company), Islamabad, Pakistan Jan 2025 – Ongoing

- Developed and deployed an end-to-end computer vision system for real-time detection and recognition of license plates, driver faces, and phone usage from camera streams, optimized for on-device inference under tight latency and compute constraints, achieving 92.4% AP@50 on phone-usage detection and an average recognition accuracy of 85%.
- Led a team of 5 developers to distill and on-premise deploy a compact Italian sentence embedding model that retained 90–95% of teacher performance on MTEB retrieval, STS, clustering, and reranking benchmarks, while significantly reducing latency and inference cost for enterprise search and RAG workloads.

Start-up Incubation, National Incubation Center, University of Balochistan June 2021 – Dec 2022

- Developed an AI-based writing tool for novel writers [Code] and raised 600,000 PKR from the Pakistan Information Technology Board in exchange for 5% equity.

Research Experience

Focus: trustworthy, uncertainty-aware vision models for Earth observation, from super-resolution to robust multimodal segmentation.

Research Assistant, Deep Learning Lab (DLL), NCAI, NUST, Pakistan Jan 2026 – Ongoing

- **Two research projects (see Research Projects):** cloud-robust SAR–optical segmentation and joint super-resolution with wildfire segmentation; manuscripts in preparation.
- **Uncertainty-aware diffusion super-resolution (manuscript in preparation):** Developed diffusion models for super-resolution that quantify their own uncertainty, using conformal prediction, wavelet transformation, and multi-modality alignment.

Research Internship, Computer Vision and Mixed Reality Lab, RheinMain University of Applied Sciences, Germany June 2024 – Aug 2024

- **State-of-the-art multi-image super-resolution (IGARSS 2026):** Reached 50.20 cPSNR and 0.9905 cSSIM on the PROBA-V dataset by integrating non-local sparse attention into the encoder of a transformer, combined with a

wavelet-enhanced loss and random sequence shuffling.

- **Single-image super-resolution of Sentinel-2 (IGARSS 2025):** Proposed an enhanced SWIN2SR transformer with non-local sparse attention and high-frequency-preserving blocks; best perceptual quality (LPIPS 0.3462, 83% user preference) and second-best PSNR (28.413 dB) on 4x super-resolution.

Research Assistant, TUKL NUST Research and Development Lab, NUST, Pakistan Sep 2023 – May 2024

- **Offline signature verification (under review, IEEE Access):** Proposed a geometry-aware hybrid CNN-Transformer with contourlet transform features, reaching state-of-the-art EERs of 6.84% (BHSig-H), 5.8% (BHSig-B), 0% (CEDAR), and 18.99% (UTSig) in writer-independent verification, with 3.6% FAR against GAN-generated fake signatures.
- Developed Generative Adversarial Networks that generate forged offline signatures to test signature-verification systems against forgery attacks.

Research Projects

Cloud-Robust SAR–Optical Semantic Segmentation from Clear-Sky Labels 2026 – Ongoing
(Supervisors: Prof. Dr. Ulrich Schwanecke, Prof. Dr. Faisal Shafait)

- Raised mIoU by 16.0 on real thick clouds (CloudSEN12) and by 1.5 over CloudSeg (official code) on its WHU-OPT-SAR protocol with one third of the parameters, using cloud-aware training of a two-stream SAR–optical network with a SAR-only head; also tested on PIE-RGB-SAR and EarthMiss over 3 seeds; manuscript in preparation.

Multi-Spectral Super-Resolution and Wildfire Segmentation (Supervisor: Prof. Dr. Faisal Shafait) 2026 – Ongoing

- Unified multi-spectral super-resolution and burnt-scar segmentation pipeline on Sentinel-2 imagery of U.S. wildfires.

Technical Skills

Computer Vision: Object detection and recognition (YOLO), semantic segmentation, super-resolution, diffusion models, GANs, vision transformers, knowledge distillation.

Programming and DL: Python, C++, SQL, PyTorch, CUDA, Hugging Face, timm, MMSegmentation, OpenCV.

Remote Sensing: Sentinel-1/2, SAR, rasterio, GDAL, GeoPandas, QGIS, Sentinel Hub, Planetary Computer.

Engineering: Linux, Git, Docker, Conda, multi-GPU training, FastAPI, AWS (EC2, S3, Lambda), MongoDB, LaTeX.

Teaching and Leadership

Guest Lecturer (6 Cr. Hrs), CS-250: Data Structure and Algorithms, NUST SEECS Oct 2026

- Taught 6 credit hours of Prof. Dr. Faisal Shafait's course during his international travel: fundamentals of C++, trees and binary trees, and binary-tree traversal algorithms.

Computer Vision Lab Instructor, Advanced AI Bootcamp, NUST Oct 2023

- Designed and delivered lectures on advanced computer vision concepts, including object detection and image segmentation.
- Guided students in implementing deep learning algorithms for image classification and object detection.

Honours and Awards

- Selected (competitive) for the IEEE GRSS IADF School on Computer Vision for Earth Observation, Italy (Sep 2026)
- DAAD grant "Super Resolution of Satellite Image": research stay in Wiesbaden, Germany (summer 2024)
- Gold Medal for academic excellence, BS Software Engineering batch of 2017, BUITEMS (Jan 2022)
- Won the Pakistan Information Technology Board Start-up competition with funding of 600,000 PKR (June 2021)
- Awarded Balochistan Talent Award for innovative product idea (Dec 2020)
- Winner of the Tech Marathon coding competition organized by USAID (Sep 2019)

Professional References

Prof. Dr. Ulrich Schwanecke
Hochschule RheinMain, Germany
ulrich.schwanecke@hs-rm.de

Prof. Dr. Faisal Shafait
SEECS, NUST, Pakistan
faisal.shafait@seecs.edu.pk